

Internet of Things

CCSW432

Course Project

SE3

Mini Arcade

MindGlow

An IoT-powered color memory game

Developed by Team BrightMinds

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1.What is MindGlow?

MindGlow is a smart and interactive color memory game designed to enhance **visual memory, concentration, and reaction speed** through an engaging sequence-based gameplay experience.

The game operates without traditional start buttons. Instead, it begins automatically when a player approaches the device using an ultrasonic proximity sensor. Once detected, the player activates the game by touching a touch sensor, providing a modern, intuitive, and seamless user interaction.

During each round, MindGlow displays a sequence of illuminated colors using LED buttons. The player must carefully observe the pattern and reproduce it by pressing the buttons in the exact same order. With every successful attempt, the game advances to the next level, where the sequence becomes progressively more challenging by increasing its length and speed.

If the player makes a mistake, the system provides visual feedback through a distinctive light pattern to indicate an error. The round then ends, and the player's final score is displayed on the LCD screen, encouraging replay and competition.

2.Project Objective

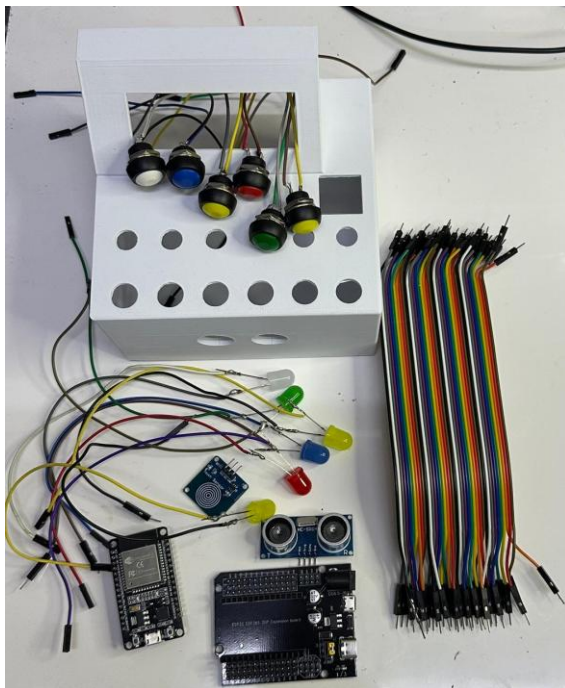
The primary objective of **MindGlow** is to improve **memory, concentration, and reaction speed** through an interactive and enjoyable gaming experience. By challenging players to observe, memorize, and accurately reproduce color sequences, the game promotes cognitive engagement in a fun and motivating way.

In addition to its cognitive goals, the project aims to demonstrate the practical application of **Internet of Things (IoT) and embedded systems technologies**. MindGlow integrates multiple sensors, visual feedback components, and wireless connectivity to create a responsive and intelligent system.

Furthermore, the project supports **performance tracking and analysis** by collecting player data through cloud services. This enables features such as score monitoring, performance comparison, and future system enhancements, highlighting how IoT solutions can combine entertainment, data collection, and real-time interaction in a single application

3. Hardware components:

- ESP32 Development Board
- ESP32 Extension Shield
- 16×2 LCD with I2C
- Touch sensor
- Ultrasonic sensor
- 6 x push buttons
- 6 x 10 mm led
- 2 x 18650 batteries 5000mAh
- 18650 batteries holder
- Dc jack



4. Software components:

- C++
- Blynk

5.How the System Works

System Workflow

- Player approaches the device → detected by ultrasonic sensor
- Player touches the touch sensor → game starts
- System generates a random color sequence
- Player inputs the sequence using buttons
- System compares input with stored pattern
- Score is displayed and stored

6.IoT & Cloud Integration

MindGlow uses IoT cloud services to monitor player performance in real time. Player scores and gameplay data can be transmitted wirelessly, enabling performance tracking, comparison, and future analytics.

7.Future Enhancements

Online leaderboard between multiple devices

Mobile app integration for player profiles

Adjustable difficulty levels based on player performance

Multiplayer or timed challenge modes

8. Conclusion

MindGlow demonstrates how IoT technologies can be applied to create interactive, engaging, and intelligent systems. By combining hardware components, embedded programming, and cloud connectivity, the project successfully delivers a fun memory-based game while showcasing practical IoT concepts learned in the course.

9. Development Steps

9.1 Add buttons



9.2 Add LED



9.3 Add Sensors and LCD



9.4 Concealing Internal Wiring

